would simply apply the historical market impact parameters and stock specific trading characteristics to compute the historical trading cost. But many active managers require a finer market impact estimate in order to differentiate cost across companies more than is possible from stock volatility and market cap. These are the active managers who are seeking to uncover superior investment opportunities through fundamental analysis using the company's balance sheet and accounting data. In these cases, it would be appropriate to correlate or define the market impact sensitivity parameter a_1 with company fundamentals to try to establish a relationship. Then, as long as we have historical company specific fundamentals, we can determine a stock specific trading cost sensitivity parameter. In Chapter 5, Estimating I-Star Model Parameters, we analyzed the error of the estimation model and showed that the error term is positively correlated with spreads, beta, and tracking error, and negatively correlated with log of market cap, and log of price. Thus these data relationships could be a very good starting point to estimate a stock specific sensitivity parameter. It is very important to point out here that this is a very time consuming and very data intensive exercise.

The question that often arises is, "Would it be better to construct stock specific market impact model?" Unfortunately, stock specific models are extremely difficult to uncover due to the nature of the data. For many stocks, price movement is dominated by the general market movement, volatility, and company news. Market impact for many stocks only has the dominating price effect if the order size is significant enough to dominate market movement and volatility. Additionally, many of the studies that uncover a stock specific relationship are not necessarily incorrect, but analysts will find it extremely difficult to establish a stock specific model across all stocks in all markets.

We encourage suspicious readers to test the validity of these findings. And to assist in this process, we outline a single stock market impact simulation exercise that can be used as the foundation for testing market impact modeling approaches later in the chapter (Market Impact Simulation).

REAL-TIME COST INDEX

A real-time cost index provides investors with the expected trading cost given actual market conditions and aggregated market imbalance. This metric combines actual volume patterns, volatility, and buying and selling pressure across all investors, and determines the fair value market price for these conditions. It is important to point out here that the derivation